Customer Behaviour Analysis

Business Problem

How can a retail company understand customer purchasing behaviour to increase revenue, improve customer retention, and optimize marketing strategies?

EDA- (Exploratory Data Analysis using Python)

Importing the libraries

```
import pandas as pd
import matplotlib.pyplot as plt
```

Importing the data

```
df = pd.read_csv('customer_shopping_behavior.csv')
df.head(4)
```

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases
2701	22	Female	T-shirt	Clothing	68.0	California	XL	Olive	Winter	3.2	No	Standard	No	36.0	Cash	Weekly
521	51	Male	Sunglasses	Accessories	84.0	South Carolina	M	White	Spring	3.9	Yes	Free Shipping	Yes	20.0	Debit Card	Quarterly
3157	18	Female	Shirt	Clothing	50.0	Montana	M	Black	Winter	3.1	No	2-Day Shipping	No	18.0	Cash	Monthly
1687	22	Male	Gloves	Accessories	75.0	Illinois	L	Red	Fall	4.2	No	Store Pickup	No	25.0	Cash	Annually

Initial Data Exploration and cleaning in jupyter notebook using pandas

- df.head()
- df.tail()
- df.describe()
- df.info()
- df.dtypes
- df.isnull().sum()
- Checked unique values
- Found inconsistencies between Item Purchased and Category
- Created a mapping dictionary
- Handling Missing Values
- identified and removed duplicates
- Converted column names

Data Analysis using SQL Server

- which category contribute most to revenue.
- Discounts may reduce profit without increasing sales.
- understanding of revenue contribution by gender segments.
- Identify high spending customers who are also discount users.
- product performance based on customer satisfaction.
- Unclear if faster shipping leads to higher spending.
- The effectiveness of subscription programs.
- Products which are overly dependent on discounts.
- Top performing products within categories.
- Relationship between loyalty and subscription
- which age group contribution most to revenue

Conclusion

The customer behaviour analysis shows that customer purchasing decisions are influenced by product category, discounts, subscription status, and purchase frequency. Loyal and repeat customers contribute a major share of the revenue, while some product categories perform significantly better than others. The analysis also highlights the importance of customer segmentation and personalized marketing strategies for improving sales and retention.

Recommendations

- Focus more on high-performing product categories to maximize revenue growth.
- Introduce loyalty and reward programs to retain repeat customers.
- Use targeted discounts instead of offering discounts to all customers.
- Promote subscription programs to increase customer engagement and recurring sales.
- Improve personalized marketing based on customer purchase behaviour and segmentation.
- Maintain proper inventory planning for high-demand products and seasonal sales.
- Enhance customer experience through better shipping services and faster delivery.

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